

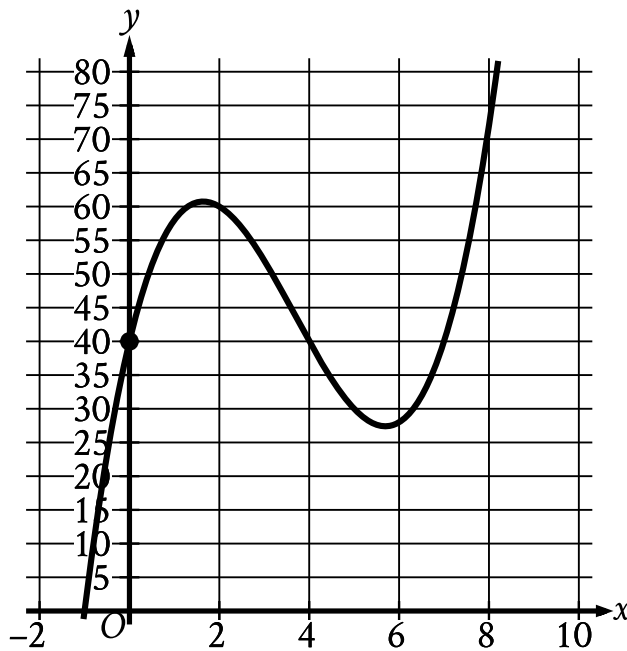
● EASY

● ADVANCED MATH

Advanced Math

08

10 questions · ~15 min



- Moving from left to right, the curve passes from quadrant 2 to quadrant 1.
- The curve has 1 relative maximum and 1 relative minimum.
- The curve passes through the following points:
 - (0 comma 40)
 - (4 comma 40)
 - (7 comma 40)

The y -intercept of the graph shown is x, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

ADVANCED MATH

$$x = 49$$

$$y = \sqrt{x} + 9$$

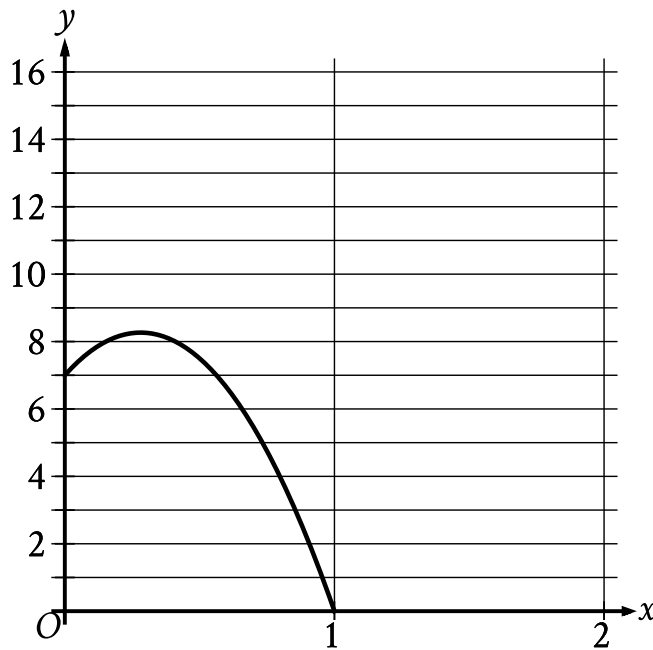
The graphs of the given equations intersect at the point x, y in the xy -plane. What is the value of y ?

- A. 16
- B. 40
- C. 81
- D. 130

The expression $2x^2 + ax$ is equivalent to $x(2x + 7)$ for some constant a . What is the value of a ?

- A. 2
- B. 3
- C. 4
- D. 7

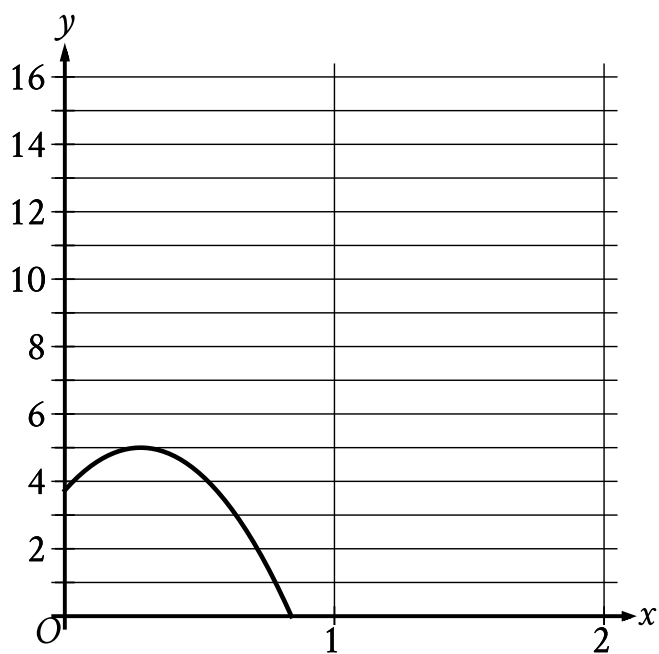
During the first part of an experiment, a ball was launched from a 7-foot-tall platform. The graph shows the height y , in feet, of the ball x seconds after it was launched during the first part of the experiment.



- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 8.3).
- The parabola passes through the following points:
 - (0 comma 7)
 - approximately (0.3 comma 8.3)
 - approximately (0.6 comma 7)

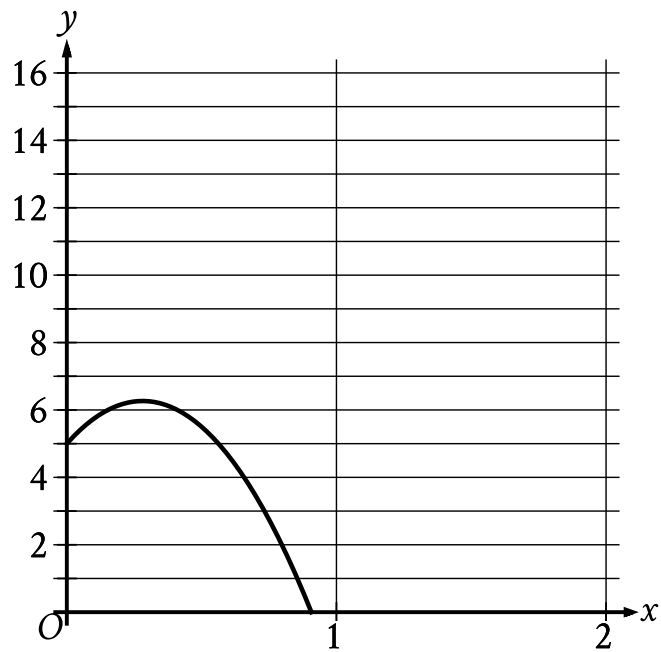
During the second part of the experiment, the ball was launched the same way, but from a platform that is 2 feet shorter than the first platform. Which of the following graphs could represent the height y , in feet, of the ball x seconds after it was launched during the second part of the experiment?

A.



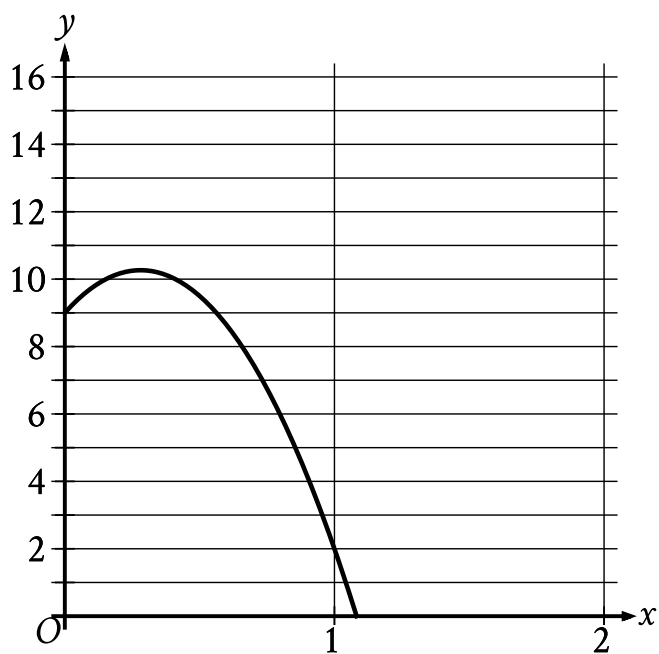
- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 5.0).
- The parabola passes through the following points:
 - approximately (0 comma 3.7)
 - approximately (0.3 comma 5.0)
 - approximately (0.6 comma 3.7)

B.



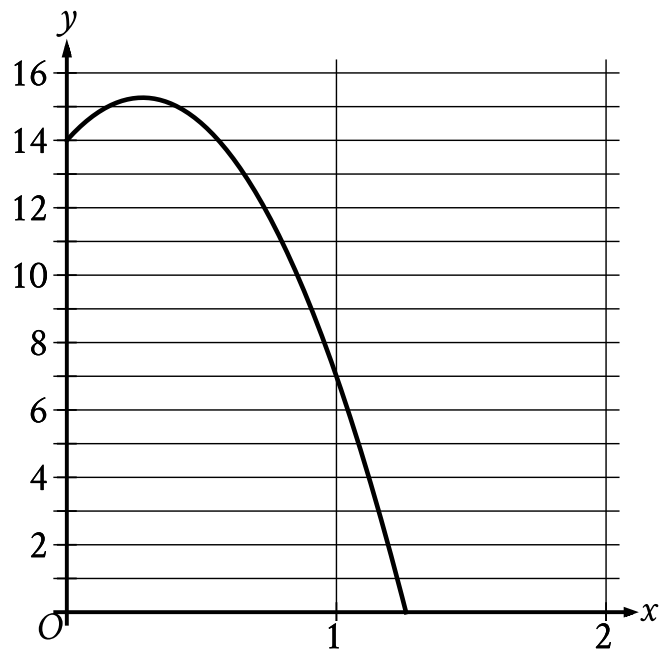
- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 6.3).
- The parabola passes through the following points:
 - (0 comma 5)
 - approximately (0.3 comma 6.3)
 - approximately (0.6 comma 5)

C.



- The parabola opens downward.
- The vertex is at the approximate point (0.3, 10.3).
- The parabola passes through the following points:
 - (0, 9)
 - approximately (0.3, 10.3)
 - approximately (0.6, 9)

D.



- The parabola opens downward.
- The vertex is at the approximate point (0.3, 15.3).
- The parabola passes through the following points:
 - (0, 14)
 - approximately (0.3, 15.3)
 - approximately (0.6, 14)

Q5

ADVANCED MATH

$$x + 2x - 5x + 9 = 0$$

What is a positive solution to the given equation?

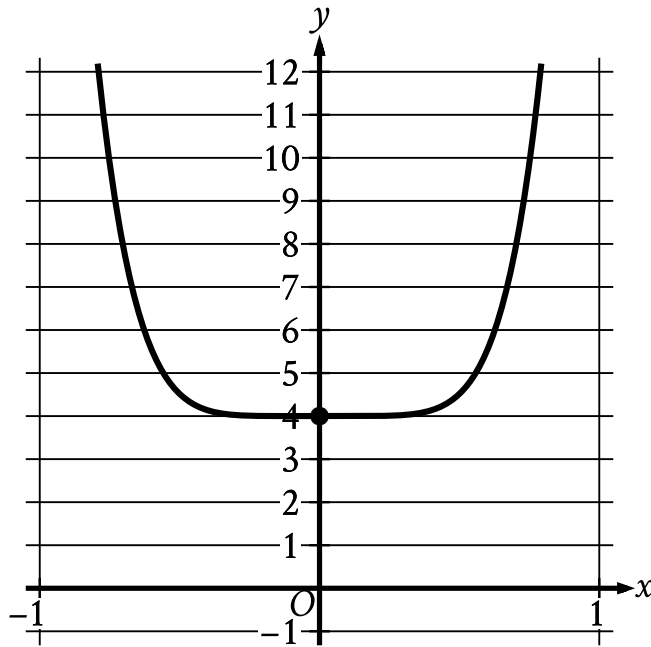
- A. 3
- B. 4
- C. 5
- D. 18

Q6

ADVANCED MATH

Which of the following is equivalent to $2(x^2 - x) + 3(x^2 - x)$?

- A. $5x^2 - 5x$
- B. $5x^2 + 5x$
- C. $5x$
- D. $5x^2$



- In quadrant 2, the curve falls to its minimum at the point (0 comma 4).
- In quadrant 1, the curve then rises.

The graph of the polynomial function f , where $y = f(x)$, is shown. The y -intercept of the graph is 0, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6

7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

ADVANCED MATH

$$7m = 2n + p$$

The given equation relates the positive numbers m , n , and p . Which equation correctly gives m in terms of n and p ?

- A. $m = \frac{2n + p}{7}$
- B. $m = 2n + p$
- C. $m = 2n + p - 7$
- D. $m = 2 - n - p - 7$

Which of the following is equivalent to $3(x+5)-6$?

- A. $3x-3$
- B. $3x-1$
- C. $3x+9$
- D. $15x-6$

$$P_t = 24.81.036^t$$

The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. According to the model, what is the best interpretation of the statement “ P_8 is approximately equal to 32.91”?

- A. In 1984, the predicted population of this country was approximately 8 million.
- B. In 1984, the predicted population of this country was approximately 32.91 million.
- C. 8 years after 1984, the predicted population of this country was approximately 32.91 million.
- D. 32.91 years after 1984, the predicted population of this country was approximately 8 million.